

Brady A. Pettengill

me@bradyap.dev ▪ (703) 980-3247 ▪ Reston, VA ▪ www.bradyap.dev

Computational and Data Sciences student at George Mason University building practical software for data analysis, physics simulation, security, and everyday use. Focused on solving real-world problems and learning new tools.

EDUCATION

B.S. Computational and Data Sciences, George Mason University
Advanced Diploma, Herndon High School

Expected Dec 2027
June 2022

PROJECTS

Adaptive Cruise Control: Basic adaptive cruise control using computer vision with Python and OpenCV, communicating with the car via CAN bus and integrating with the existing cruise control through a PID controller.

Messaging App: Desktop chat platform with client-server functionality, written in Java using Swing and MySQL.

N-body Simulation: Gravitational N-body sim written in C++ with OpenMP, std::thread, and serial backends to benchmark CPU parallelization. Exposed to Python via pybind11 for real-time visualization.

Utility Website: A collection of Node.js/Express + MongoDB web apps with a Bootstrap/HTML frontend, including a URL shortener and pastebin exposed via REST APIs and the UI.

EXPERIENCE

George Mason Cyber Team
CTF Competitor

Sep. 2023 – Present
Fairfax, VA

- Compete in NCL and collegiate CTFs, focusing on web, crypto, and reverse-engineering challenges.
- Apply tools such as Burp Suite, Wireshark, Nmap, Ghidra, and GDB to identify and exploit vulnerabilities.
- Collaborate with teammates to solve real-world-style cybersecurity problems within strict time limits.

State Farm Insurance
Account Manager

May 2024 – Sep. 2025
Vienna, VA

- Drove sales growth by identifying customer needs and presenting tailored insurance solutions.
- Used CRM workflows and Excel-based tracking and reporting to manage a high-volume account pipeline.

Nando's PERi-PERi
Restaurant Manager

Dec. 2021 – May 2024
Reston, VA

- Oversaw financial performance, inventory management, marketing, and hiring/scheduling staff.

FIRST Robotics FTC Team
Software Team Lead

Aug 2021 – June 2022
Herndon, VA

- Organized our team to successfully design, implement, and test software for our robot.
- Developed software for both manual control and autonomous operation.

Herndon High School AFA CyberPatriot Team
Founder and President

Sep. 2020 – June 2022
Herndon, VA

- Founded HHS CyberPatriot team "CyberHornet," reaching national semifinals in both years of competition.
- Developed our Linux sub-team, introducing members to the use and hardening of Linux-based systems.
- Worked extensively with bash scripting to automate standard competition procedures.

SKILLS

Languages: Python, C, C++, R, MATLAB, JavaScript, Java, SQL, Swift, Bash

Libraries: NumPy, Pandas, Scikit-learn, OpenCV, MongoDB, Node.js, Express

Tools/Platforms: Git, Linux, Docker, NGINX, Cloudflare, Wireshark, Ghidra, GDB, Nmap

Soft Skills: Team collaboration, problem-solving, adaptability, technical communication, leadership